LAKSHMI NARAIN COLLEGE OF TECHNOLOGY (MCA), BHOPAL

MCA – I SEMESTER

Information Technology

MCA-104

Assignment

UNIT-II (CO-II)

Case Study 1: Information Security and Mobile Commerce in an Online Shopping Application

An Indian e-commerce company wants to develop a mobile shopping application that allows customers to register, log in, browse products, make online payments, and track their orders. The application stores sensitive information, including customer details, passwords, payment records, and transaction histories.

The company wants to ensure secure transactions, protect customer information from cyberattacks, and promote its products through digital marketing. It plans to use encryption, digital signatures, firewalls, user authentication, and mobile commerce technologies to build a secure and reliable application.

Questions:

As a system designer, explain the complete working of this system. Your answer must cover the following points:

i. Explain the major information security threats to the application, including malicious programs such as viruses & worms.

ii. Explain how cryptography & digital signatures can protect the application and its transactions.

iii. Describe the role of security awareness and security policies in protecting customer data.

iv. Explain the concept and growth of mobile commerce, its major technologies, and its development in India.

v. Recommend suitable digital marketing strategies for promoting the application.

OR

Case Study 2: Secure Mobile Banking and Digital Payment System

A bank wants to introduce a mobile banking application that enables customers to check account balances, transfer money, pay bills, and make merchant payments using smartphones. The bank plans to support digital payment services such as UPI and mobile wallets.

Since financial transactions involve sensitive information, the bank must protect its systems against malware, phishing, identity theft, unauthorized access, and fraudulent transactions. It also wants to increase the adoption of mobile banking through customer awareness campaigns and digital marketing.

Questions:

As a system designer, explain the complete working of this system. Your answer must cover the following points:

i. Identify the major security threats associated with mobile banking.

ii. Explain how firewalls and user identification and authentication can help secure mobile banking transactions.

iii. Discuss the importance of multi-factor authentication and safe mobile banking practices.

iv. Explain the technologies used in mobile banking and digital payments of mobile commerce services.

v. Suggest suitable digital marketing strategies to increase the adoption of mobile banking while maintaining customer trust, privacy, and security.

Note:

i. Attempt any one of the above case studies.

ii. Give suitable examples and diagrams wherever necessary.

iii. Make neat and clean assignment file clearly mentioning your name, roll number, section, and Subject code with subject name at the front page.